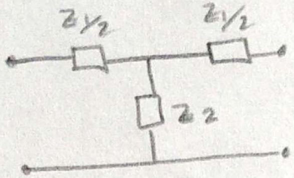
7) $f_c = 1K$ con $600\,\Omega$, $f_{att\,\infty}$ en $1{,}05K$, secciones de $m=0{,}6$ IN y OUT.

Z_L

una red T prueba

- $f_c = 1kHz \Rightarrow \omega_c = 2\pi f_c = 2000\pi$
- $f_0 = 1K05Hz \Rightarrow \omega_0 = 2\pi f_0 = 2100\pi$



- $Z_1 = SL$
- $Z_2 = \frac{1}{SC}$
- $R_0 = \sqrt{\frac{L}{C}} = (600\,\Omega)$ (I)
- $Z_{0T} = \sqrt{\frac{L}{C}}\sqrt{1 - \frac{\omega^2 LC}{4}}$

(corte donde $Z_{0T} = 0$)

$1 - \frac{\omega_c^2 LC}{4} = 0$

$1 = \frac{\omega_c^2 LC}{4}$

$4 = \omega_c^2 LC$

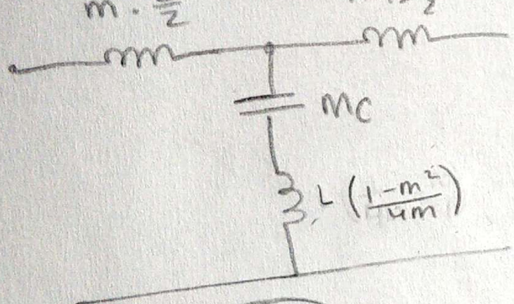
$\omega_c = \sqrt{\frac{4}{LC}}$

frec de corte

(I) donde $L = C \cdot 360K$.

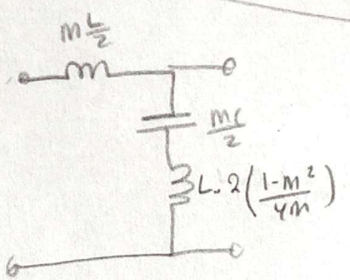
$\omega_c = 2000\pi = \frac{2}{\sqrt{LC}}$

$2000\pi = \frac{2}{\sqrt{C \cdot 360K \cdot C}} \Rightarrow \boxed{C = 530{,}5nF} \Rightarrow \boxed{L = 190{,}9mHy}$



$\Rightarrow$

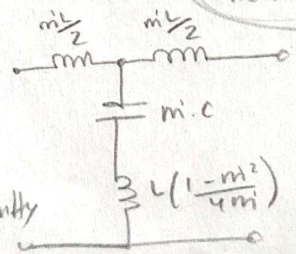
$m = 0{,}6$



- $\frac{m \cdot L}{2} = 57{,}27mHy$
- $\frac{mC}{2} = 159{,}15nF$
- $L2\left(\frac{1-m^2}{4m}\right) = 101{,}86mHy$
- $\omega_\infty = \frac{\omega_c}{\sqrt{1-m^2}} = 2500\pi \neq 2100\pi$

tengo que "derivar"

$m = \sqrt{1 - \left(\frac{f_c}{f_\infty}\right)^2} = 0{,}305$



- $\frac{m'L}{2} = 29{,}11mHy$
- $m'C = 161{,}8nF$
- $L\left(\frac{1-m'^2}{4m'}\right) = 141{,}92mHy$

Escaneado con CamScanner